CorrCal Reading Plan

Statistics, numerics, symmetry, and geometry for gpu_corrcal

Each phase ends in a concrete computation on your own array, so the reading gets tested against gpu_corrcal rather than sitting in the abstract. Check chapter numbers against your editions, since a few are from memory of the tables of contents.

Phase 1: How the likelihood becomes Ax = b

The goal is to derive CorrCal from first principles and see exactly where the linear systems appear.

Readings

- **Shewchuk**, *An Introduction to the Conjugate Gradient Method Without the Agonizing Pain*: §3, §7–9, and §12–14. The quadratic form, CG, preconditioning, CG on the normal equations, and nonlinear CG.
- **Bishop**, *Pattern Recognition and Machine Learning*: §2.3 and §3.1, on marginalizing a linear-Gaussian latent variable and on ML as least squares.
- **Hogg, Bovy & Lang** (arXiv:1008.4686): GLS with a full covariance.
- **Liu et al. 2010** (arXiv:1001.5268): logcal and lincal as linear least squares.
- **Rasmussen & Williams**, *Gaussian Processes for Machine Learning*: ch. 2, §5.4, and Appendix A.3, on the marginal likelihood, its gradients, Woodbury, and the determinant lemma.
- **Sievers 2017** (arXiv:1701.01860): read this last, once the pieces are in place.

Probe

On paper, derive the CorrCal χ^2 and its gradient with respect to the gains. Write out the Gauss-Newton normal equations for a small redundant array. Then check the analytic gradient against finite differences in gpu_corrcal.

Phase 2: Estimation theory and conditioning

The goal is to understand how good the estimator is and why CG converges at the rate it does.

Readings

- **Kay**, *Fundamentals of Statistical Signal Processing, Vol. I: Estimation Theory*: ch. 3 on Cramér-Rao, ch. 4 on the linear model, ch. 7 on MLE, and ch. 15 on complex data. Try the McGill library.
- **Nocedal & Wright**, *Numerical Optimization*: ch. 5 on CG and ch. 10 on nonlinear least squares.
- **Boyd & Vandenberghe**, *Convex Optimization*: §7.1 and Appendix C.
- **Tegmark 1997** (astro-ph/9611130): map-making as preconditioned CG on a GLS system.

Probe

Compute the Fisher matrix for a CHORD-like layout and plot its eigenspectrum. Identify the near-zero eigenvalues, and relate the condition number of the non-degenerate block to the CG iteration counts you actually see. Compare the Cramér-Rao bounds on the gains with the scatter gpu_corrcal produces in simulation.

Phase 3: Groups and representations

Three facts you already rely on are really representation theory:

- **Grid redundancy.** CHORD's regular grid is invariant under translations, so the redundant covariance is block-circulant. Circulant matrices are diagonalized by the DFT because the Fourier modes are the characters of the cyclic group, and this is what makes FFT-based preconditioners work.
- **Calibration degeneracies.** The degeneracies of redundant calibration (overall amplitude, overall phase, and two phase tilts) form an abelian Lie group acting on the gains. Its Lie algebra spans the flat directions of the likelihood.
- **Sky rotation.** For a drift-scan telescope, Earth's rotation is a symmetry. The m-mode formalism is Fourier analysis on U(1). An isotropic sky covariance is diagonal in spherical harmonics, which is Schur's lemma applied to SO(3).

The readings for this phase:

- **Zee**, *Group Theory in a Nutshell for Physicists*: the early parts on finite groups, representations, Schur's lemma, and characters, then the rotation group and Lie algebras.
- **Serre**, *Linear Representations of Finite Groups*, Part I: short, rigorous character theory. Read it after Zee's version.
- **Diaconis**, *Group Representations in Probability and Statistics* (free online): the representation basics and the spectral analysis chapter, which apply representation theory to statistical data.
- **Shaw et al. 2014** (arXiv:1302.0327) and **2015** (arXiv:1401.2095): the m-mode formalism for transit telescopes.

Probes

- Prove that the DFT diagonalizes circulant matrices using characters of $\mathbb{Z}_n$, then extend the argument to $\mathbb{Z}_n \times \mathbb{Z}_m$ for the grid.
- Build a circulant preconditioner for the redundant block and measure its effect on CG iteration counts in gpu_corrcal.
- Write down the degeneracy group, compute its generators, and verify that they span the numerical null space you found in Phase 2.

Phase 4: Geometry

The goal is to recast calibration as optimization on a curved space.

Readings

- **Zee**, *Einstein Gravity in a Nutshell*: Part I for tensors, metrics, connections, and curvature. Part II for action, symmetry, and Noether.
- **Carroll**, *Spacetime and Geometry*: ch. 2–3, on manifolds and curvature, more carefully than Zee.
- **Amari**, *Information Geometry and Its Applications*: Part I, on the Fisher metric and dual connections.
- **Absil, Mahony & Sepulchre**, *Optimization Algorithms on Matrix Manifolds* (free online): ch. 3–6, on quotient manifolds, the Riemannian gradient and Hessian, and Newton's method.

Probe

Formulate gain calibration on the quotient manifold of gains modulo the Phase 3 degeneracy group. Compare a natural-gradient step with preconditioned CG in the Fisher metric, and check whether fixing the gauge geometrically beats adding penalty terms.

Order and thesis payoff

Go 1 → 2 → 4, with Phase 3 running alongside and feeding into both 2 and 4.

Phases 1 and 2 give you the statistical foundation chapter of the thesis and the justification that the estimator is near-optimal.

The circulant preconditioner from Phase 3 is a concrete performance result for gpu_corrcal. The symmetry treatment of the degeneracies gives you a clean way to present gauge fixing.

Phase 4 is the most open-ended. If the quotient-manifold formulation improves convergence, it is a publishable methods contribution. If it does not, it still gives you a sharper explanation of why the degenerate directions behave the way they do. Either result is worth having.